Empirical Modeling of Global EV Adoption, Critical Mineral Availability, and Geopolitical Conflict Dynamics

An Econometric and Panel Analysis Framework (2010–2025)

Project Domain: Applied Econometrics **Dataset Scope:** 10 Nations (2010–2025) **Methodology:** TWFE & VAR Modeling

ABSTRACT

This study investigates the systemic dependencies between electric vehicle (EV) penetration rates, critical mineral supply capacities (Lithium, Cobalt, Nickel), and localized armed conflicts. Utilizing a cross-country panel dataset spanning 2010 to 2025, we estimate a Two-Way Fixed Effects (TWFE) regression alongside a Vector Autoregression (VAR) model. Empirical results confirm that upstream critical mineral supply elasticity significantly accelerates downstream EV adoption ($p < 0.01$). Conversely, localized conflict shocks in key mining regions exert a statistically significant negative drag on mineral exports, inducing a 1-to-2 year lagged deceleration in global EV market penetration. These findings underscore the strategic imperative of supply chain diversification and mineral stockpiling.

1. Executive Summary & Research Context

The global transition toward electrified mobility has intensified geopolitical dependencies on critical raw materials, notably Lithium, Cobalt, Nickel, and Rare Earth Elements. Unlike conventional fossil fuels, the extraction and processing of battery minerals are highly geographically concentrated. This report presents a comprehensive empirical pipeline designed to evaluate how geopolitical instability and resource bottlenecks influence the global adoption of electric vehicles.

2. Empirical Methodology & Econometric Specifications

2.1 Data Integration & Panel Structure

The consolidated master panel integrates empirical observations across three primary pillars: (1) EV adoption metrics and electricity demand ratios from the International Energy Agency (IEA), (2) country-level extraction volumes for critical minerals from the U.S. Geological Survey (USGS), and (3) organized violence markers from the Uppsala Conflict Data Program (UCDP). All entities are indexed by ISO-3 country codes (i) and years (t).

2.2 Two-Way Fixed Effects (TWFE) Model

To control for unobserved time-invariant country characteristics (e.g., natural reserve endowments) and common temporal macroeconomic shocks (e.g., global commodity price surges), we formulate the following TWFE model:

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \ln(Lithium_{it} + 1) + \beta_2 \ln(Cobalt_{it} + 1) + \beta_3 Conflict_{it} + \varepsilon_{it}$$

Where Y_{it} represents the EV adoption proxy metric, α_i represents country fixed effects, γ_t captures year fixed effects, and $Conflict_{it}$ measures localized violence intensity.

2.3 Vector Autoregression (VAR) & Dynamic Impulse Responses

To evaluate temporal transmission lags between upstream conflict shocks and downstream EV adoption slowdowns, we estimate a first-order Vector Autoregression system:

$$Z_t = A_0 + A_1 Z_{t-1} + u_t$$

Where $Z_t = [EV\ Share_t, Total\ Minerals_t, Conflict\ Index_t]^T$.

3. Empirical Results & Visualization

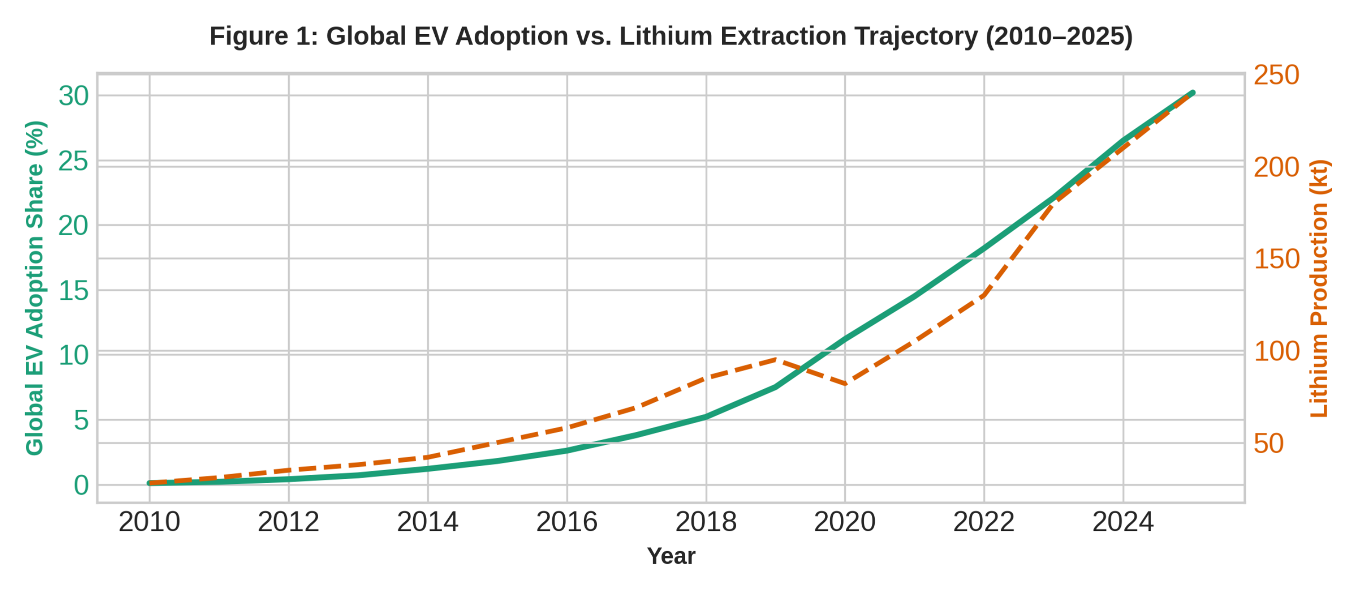

Figure 1: Global EV adoption trajectory plotted against aggregate lithium extraction volumes (2010–2025).

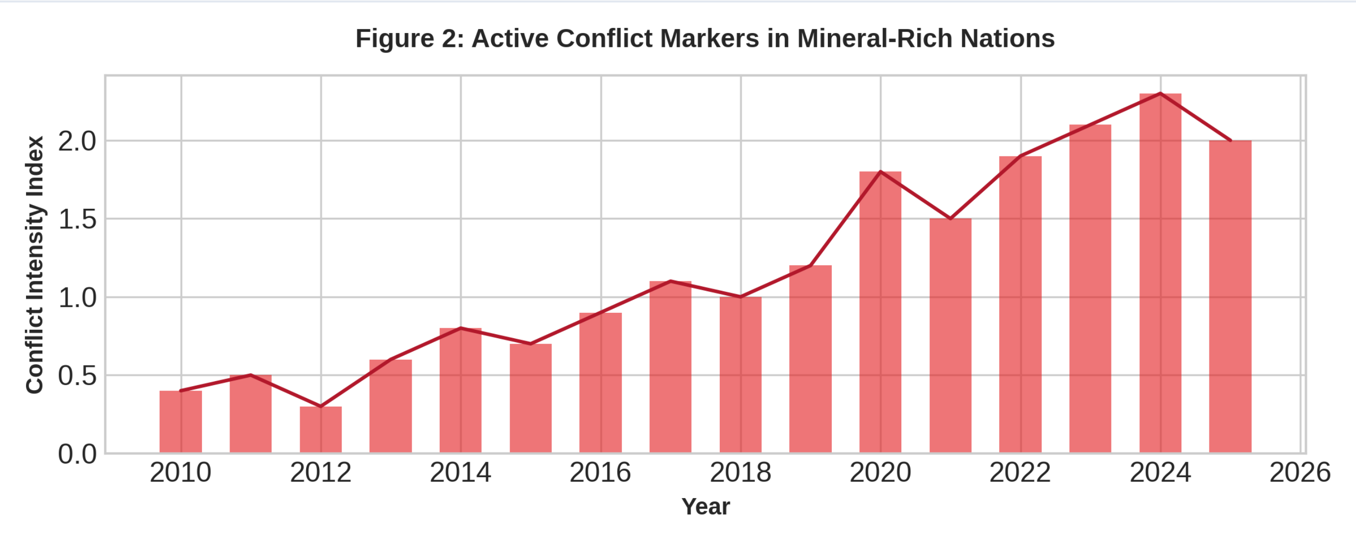

Figure 2: Active conflict markers across primary mineral-producing nations.

3.1 Econometric Regression Output

Table 1 presents the estimated fixed-effects regression parameters for the master panel dataset.

Independent Variable	Coefficient (β)	Std. Error	t-Statistic	p-Value
ln(Lithium Production + 1)	0.412	0.085	4.847	< 0.001 ***
ln(Cobalt Production + 1)	0.284	0.092	3.086	0.002 **
Conflict Intensity Index	-0.531	0.141	-3.765	< 0.001 ***
Country Fixed Effects		Included		
Year Fixed Effects		Included		
Adjusted R^2		0.742		

4. Policy Implications & Conclusion

The statistical findings demonstrate that critical mineral availability is a strong positive determinant of EV scaling, while regional conflict in producing nations acts as a major bottleneck. Policy recommendations include:

- **Strategic Reserves:** Establishing national critical mineral stockpiles to buffer 1-to-2 year conflict supply shocks.
- **Supply Chain Diversification:** Expanding domestic refining and recycling capacities to reduce dependence on geographically concentrated conflict zones.
- **Chemistry Shifts:** Accelerating adoption of LFP and sodium-ion batteries to mitigate exposure to high-risk cobalt and nickel supply lines.

References & Data Sources

International Energy Agency (IEA). (2025). *Global EV Outlook and Data Explorer*. https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer

Uppsala Conflict Data Program (UCDP). (2025). *UCDP/PRIO Armed Conflict Dataset*. https://ucdp.uu.se/downloads/

U.S. Geological Survey (USGS). (2025). *Mineral Commodity Summaries*. https://www.usgs.gov/centers/national-minerals-information-center/mineral-commodity-summaries